

Bayesian Bootstrap for the Transition Matrix of Finite State-Space Markov Chains

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GRADUATE STUDENT SEMINAR

September, 2021

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Introduction

This talk deals with the following statistical problem:

The Problem

Given a Markov chain $(X_n)_{n \geq 0}$ with an *unknown* transition matrix P , estimate P on the basis of an observed path $(x_0, x_1, \dots, x_n)$.

The structure of the talk goes as follows:

- ① Asymptotic approach to the problem
- ② Frequentist Bootstrap approach to the problem
- ③ Bayesian Bootstrap approach to the problem.

The first two points will be presented very briefly, with the intention of serving as preparatory material for the Bayesian Bootstrap approach.

Different approaches to our problem

We have four main approaches for the problem of interest:

- **Asymptotic:** take a large number of samples and estimate the transition matrix using the sample distribution.
 - ① Frequentist: find the maximum likelihood estimator (MLE) $\hat{P}$ of P
 - ② Bayesian: assign appropriate prior probabilities to the transitions and calculate the posterior distribution of the transition matrix.
- **Bootstrap:** use the initial sample path to get the bootstrap distribution via resampling and simulation methods.
 - ① Frequentist: parametric bootstrap paradigm
 - ② Bayesian: simulate standardized exponential distributions and use them to estimate the transition probabilities.

We will discuss very briefly the asymptotic and frequentist bootstrap approaches, with our emphasis given on the Bayesian bootstrap framework.

Asymptotic frequentist approach

- Let $\{X_n : n \in \mathbb{N}\}$ be a time-homogeneous Markov chain with finite state space $S = \{1, \dots, m\}$ and transition probability matrix $P = (p_{ij})_{i,j \in S}$.
- *Data*: a sample path $(x_0, x_1, \dots, x_n)$ of the chain
- *Goal*: estimate all m^2 elements of P based on the path
- Parametric space 1: The *natural parametric space* is the Cartesian product $\Theta = \Theta_1 \times \Theta_2 \times \dots \times \Theta_m$, where each Θ_i is the probability simplex

$$\Theta_i = \left\{ (p_{i1}, \dots, p_{im}) : 0 \leq p_{i1}, \dots, p_{im} \leq 1, \sum_{k=1}^m p_{ik} = 1 \right\} \subseteq [0, 1]^m.$$

- Parametric space 2: The *minimal parametric space* is the Cartesian product $\Theta^* = \Theta_1^* \times \Theta_2^* \times \dots \times \Theta_m^*$, where each Θ_i^* is defined by

$$\Theta_i^* = \left\{ (p_{i1}, \dots, p_{i,i-1}, p_{i,i+1}, \dots, p_{im}) : \sum_{k \in S_i} p_{ik} \leq 1 \right\} \subseteq [0, 1]^{m-1},$$

where $S_i := S \setminus \{i\} = \{1, \dots, i-1, i+1, \dots, m\}$.

Likelihood

The likelihood $L(\lambda, P)$ is given by

$$\begin{aligned}
 L(\lambda, P) &= \mathbb{P}(X_0 = x_0, X_1 = x_1, \dots, X_n = x_n) \\
 &= \lambda_{x_0} \cdot \mathbb{P}(X_1 = x_1 \mid X_0 = x_0) \dots \mathbb{P}(X_n = x_n \mid X_{n-1} = x_{n-1}, \dots, X_0 = x_0) \\
 &= \lambda_{x_0} \cdot \mathbb{P}(X_1 = x_1 \mid X_0 = x_0) \dots \mathbb{P}(X_n = x_n \mid X_{n-1} = x_{n-1}) \\
 &= \lambda_{x_0} \cdot \prod_{t=1}^n p_{x_{t-1}x_t} = \lambda_{x_0} \cdot \prod_{i \in S} \prod_{j \in S} p_{ij}^{n_{ij}(n)}.
 \end{aligned}$$

We assume that the chain starts from a fixed initial state, so we will work conditioning on X_0 . Thus, the likelihood becomes

$$L_n(P) = \mathbb{P}(X_1, \dots, X_n; P \mid x_0) = \prod_{i \in S} \prod_{j \in S} p_{ij}^{n_{ij}(n)} = \prod_{i=1}^m \prod_{j=1}^m p_{ij}^{n_{ij}(n)}, \quad (1)$$

where the counting processes $(n_{ij}(n))_{n \in \mathbb{N}}$ and $(n_i(n))_{n \in \mathbb{N}}$ are defined next.

Maximum Likelihood Estimation

The *counting processes* $(n_{ij}(n))_{n \in \mathbb{N}}$ and $(n_i(n))_{n \in \mathbb{N}}$ are defined by

$$n_{ij}(n) = \sum_{k=0}^{n-1} \mathbb{1}_{\{X_k=i, X_{k+1}=j\}} \quad \text{and} \quad n_i(n) = \sum_{k=0}^{n-1} \mathbb{1}_{\{X_k=i\}}.$$

We will also be writing n_{ij} and n_i instead of $n_{ij}(n)$ and $n_i(n)$, respectively.

Theorem (M.L.E. of the Transition Matrix)

Let $(X_n)_{n \in \mathbb{N}}$ be a discrete-time homogeneous Markov chain with finite state-space S , initial distribution λ and transition matrix P . Let $\{x_0, x_1, \dots, x_n\}$ be an observed realization of the chain. If $\hat{P}$ denotes the maximum likelihood estimator of the transition matrix based on the given path at time n , then we have that $\hat{P} = (\hat{p}_{ij})_{i,j \in S}$, where

$$\hat{p}_{ij} = \begin{cases} \frac{n_{ij}}{n_i}, & \text{if } n_i > 0 \\ 0, & \text{if } n_i = 0 \end{cases}. \quad (2)$$

Asymptotic Behavior

Theorem (Consistency of the M.L.E.)

For every $i, j \in S$, we have that $\hat{p}_{ij} \xrightarrow{a.s.} p_{ij}$, i.e. the maximum likelihood estimator $\hat{P} = (\hat{p}_{ij})_{i,j \in S}$ is strongly consistent for the transition matrix P .

Theorem

Let $\pi = (\pi_1, \dots, \pi_m)$ denote the stationary distribution of $(X_n)_{n \in \mathbb{N}}$. The maximum likelihood estimator matrix $\hat{P}$ is asymptotically normal, i.e.

$$\sqrt{n}(\hat{P} - P) \xrightarrow{d} \mathcal{N}_{m^2}(0, \text{diag}(\pi_i^{-1} \Lambda_i)),$$

where each Λ_i is the covariance matrix

$$\Lambda_i = \begin{pmatrix} p_{i1}(1 - p_{i1}) & -p_{i1}p_{i2} & \cdots & -p_{i1}p_{im} \\ \vdots & \vdots & \ddots & \vdots \\ -p_{im}p_{i1} & -p_{im}p_{i2} & \cdots & p_{im}(1 - p_{im}) \end{pmatrix}.$$

Different cases of Bootstrap

There are 4 main different cases of Bootstrap:

- Frequentist Bootstrap for independent data
- Frequentist Bootstrap for dependent data
- Bayesian Bootstrap for independent data
- Bayesian Bootstrap for dependent data

The Bootstrap method was initially developed for independent data (e.g. for iid random variables). Here, we will study the Bootstrap method when we have the simplest dependence structure, that of a Markov chain.

Frequentist Bootstrap - The Idea

The typical Nonparametric Bootstrap scheme is the following:

- ① Take a sample that is **representative of the population**.
- ② Calculate the statistic of interest in this sample.
- ③ Simulate **with replacement** from the initial sample.
- ④ Calculate the statistic of interest from this (bootstrapped) sample.
- ⑤ Iterate the above procedure many times and gather the estimations.
- ⑥ Take the average of these estimates as an estimate of the statistic of interest.

Step 3 can happen in a different way.

- Instead of keeping track of the observations themselves, we can count **how many times** each observation occurs and construct weights.
- The distribution of the weights is multinomial with equal probabilities.
- Simulate from the **multinomial distribution** and use the weighted averages as a bootstrap value (if the statistic of interest is the sample mean).

Parametric Bootstrap for Markov Chains

The goal now is to estimate the sampling distribution of $\sqrt{n}(\hat{P}_n - P)$. The steps of the parametric bootstrap algorithm are summarized as follows:

- ① Find an estimate of P . Here we will use the mle $\hat{P}_n$.
- ② Use $\hat{P}_n$ as a transition matrix and generate a path $(X_0^*, X_1^*, \dots, X_{N_n}^*)$.
- ③ Using the plug-in principle, approximate the sample distribution of $R(\underline{X}, P)$ by the distribution of $R^* := R(\underline{X}^*, \hat{P}_n)$. That is, find the mle $\tilde{P}_n$ as if the matrix $\hat{P}_n$ that generated the data was the unknown transition matrix, and study the distribution of $\sqrt{N_n}(\tilde{P}_n - \hat{P}_n)$.

Asymptotic Behavior

Using the Lindeberg-Feller Central Limit Theorem, it can be proved that the asymptotic maximum likelihood estimator and the parametric bootstrap estimator have the same asymptotic distributions.

Theorem

Let $(X_n)_{n \geq 0}$ be an aperiodic, irreducible, time-homogeneous Markov chain with transition matrix P and finite state-space $S = \{1, \dots, m\}$. Let $\hat{P}_n$ be its maximum likelihood estimator based on an observed path $\underline{x} = (x_0, x_1, \dots, x_n)$. For $\tilde{P}_n$ as described above, we have the asymptotic result

$$\sqrt{N_n}(\tilde{P}_n - \hat{P}_n) \xrightarrow{d} \mathcal{N}_{m^2}(0, \Sigma), \quad (3)$$

where Σ is the covariance matrix of the asymptotic distribution defined in Theorem 3.

Bayesian Bootstrap - The Idea

Suppose the statistic of interest is the sample mean.

- The frequentist bootstrap procedure yields the bootstrap value $\sum_{i=1}^n w_i x_i$ at each step, where $w_i \sim \text{Multinomial}(1, 1/n, \dots, 1/n)$.
- Problem: the discrete nature of the Multinomial distribution leads to **non-smooth** estimators.

The Bayesian Bootstrap solves the above problem by using a continuous version of the weights' distribution, the **Dirichlet** distribution. We have the following:

- ① Take a sample $\{x_1, \dots, x_n\}$ and calculate the statistic of interest.
- ② Generate the weights $w_i \sim \text{Dir}(1, \dots, 1)$, $i = 1, \dots, n$.
- ③ Take the bootstrap value $\sum_{i=1}^n w_i x_i$.
- ④ Iterate the previous steps and take the mean of all the bootstrap values.

The above procedure leads to **smoother** estimators. Note that now the weights are random variables.

Dirichlet distribution

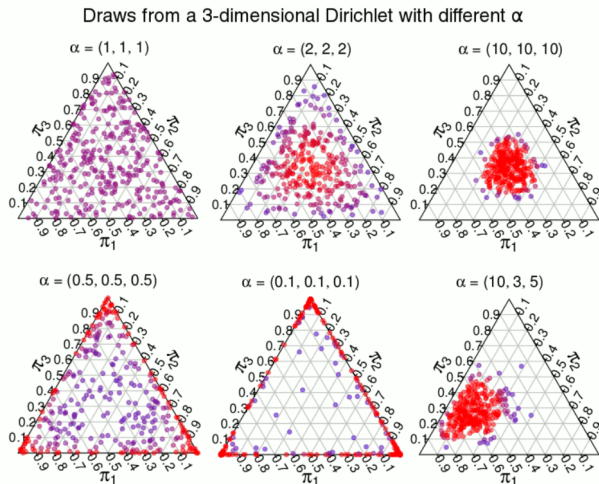


Figure: Samples from Dirichlet distributions with different parameters.

Source: Rasmus Bååth's Research Blog (URL: <http://sumsar.net/blog/2015/04/the-non-parametric-bootstrap-as-a-bayesian-model/>)

Beta, Dirichlet & Matrix-Beta distributions

- *Beta*:

$$X \sim \text{Beta}(p, q) \Leftrightarrow f_X(x) = \frac{\Gamma(p+q)}{\Gamma(p)\Gamma(q)} x^{p-1} (1-x)^{q-1}, \quad x \in (0, 1)$$

- *Dirichlet*:

$$X \sim \text{Dir}(c_1, \dots, c_m) \Leftrightarrow f_X(x_1, \dots, x_m) = \frac{\Gamma\left(\sum_{i=1}^m c_i\right)}{\prod_{i=1}^m \Gamma(c_i)} \prod_{i=1}^m x_i^{c_i-1},$$

for $(x_1, \dots, x_m)$ in the $(m-1)$ -probability simplex. The Dirichlet distribution is the multidimensional analogue of the Beta distribution.

- *Matrix-Beta*: the distribution of a matrix whose rows are independent, each one following a Dirichlet distribution.

Bayesian Bootstrap for Markov Chains - Notation

Let B_{ij} denote the set $B_{ij} = \left\{ \sum_{\ell=1}^j n_{i(\ell-1)} + 1, \dots, \sum_{\ell=1}^j n_{i\ell} \right\}$, $i, j \in S$, where we define $n_{i0} = 0$ for all $i \in S$. Note that each B_{ij} can be written as

$$B_{ij} = \{n_{i1} + \dots + n_{i(j-1)} + 1, \dots, n_{i1} + \dots + n_{i(j-1)} + n_{ij}\},$$

from which we conclude that each B_{ij} has n_{ij} elements. Hence, the maximum likelihood estimator (1) can be written as

$$\hat{p}_{ij} = \frac{n_{ij}}{n_i} = \frac{\sum_{t \in B_{ij}} 1}{\sum_{t=1}^{n_i} 1}, \quad (4)$$

which works as a useful connection between the asymptotic and the Bayesian bootstrap approach. We will use this connection to define the Bayesian bootstrap estimators for the transition probabilities.

Example

We will give an example in order to understand the notation better. Let $N = 20$ and suppose we have a Markov chain path such that

$$n_{11} = 2, n_{12} = 2, n_{13} = 2, n_{21} = 1, n_{22} = 1, n_{23} = 4, n_{31} = 2, n_{32} = 4, n_{33} = 1.$$

Then we have that $n_1 = n_{11} + n_{12} + n_{13} = 6$, $n_2 = n_{21} + n_{22} + n_{23} = 6$ and $n_3 = n_{31} + n_{32} + n_{33} = 7$. The sets B_{ij} are defined as:

$$B_{11} = \{n_{10} + 1, \dots, n_{11}\} = \{1, 2\} \rightsquigarrow |B_{11}| = n_{11} = 2$$

$$B_{12} = \{n_{10} + n_{11} + 1, \dots, n_{11} + n_{12}\} = \{3, 4\} \rightsquigarrow |B_{12}| = n_{12} = 2$$

$$\vdots$$

$$B_{32} = \{n_{30} + n_{31} + 1, \dots, n_{31} + n_{32}\} = \{3, 4, 5, 6\} \rightsquigarrow |B_{32}| = n_{32} = 4$$

$$B_{33} = \{n_{30} + n_{31} + n_{32} + 1, \dots, n_{31} + n_{32} + n_{33}\} = \{7\} \rightsquigarrow |B_{33}| = n_{33} = 1$$

Notice that sets $B_{i1}, B_{i2}, \dots, B_{im}$ *partition* the set $\{1, 2, \dots, n_i\}$, for all $i \in [m]$.

Bayesian Bootstrap Estimator

We will use an estimator similar to the M.L.E. (4), but will now include simulation.

Definition (Bayesian Bootstrap Estimator)

For every $i, j \in S = [m]$, the Bayesian bootstrap estimator of p_{ij} is defined to be

$$\hat{p}_n^*(i, j) := \frac{\sum_{t \in B_{ij}} Z_{it}}{\sum_{t=1}^{n_i} Z_{it}}, \quad (5)$$

where $Z_{it} \sim \text{Exp}(1)$ are iid random variables, for $i = 1, \dots, m$, $t = 1, \dots, n_i$.

Let $P = (P_1 \cdot P_2 \cdots P_m)^T$ denote the unknown transition matrix P and $\hat{P}_n^* = (\hat{p}_n^*(i, j))$ the Bayesian bootstrap estimator. It can be shown that the joint distribution of each row of $\hat{P}_n^*$ is Dirichlet with parameters involving the quantities n_{ij} . In particular, $\hat{P}_{i\cdot}^* \sim \text{Dir}(n_{i1}, \dots, n_{im})$. Thus the joint distribution of the Bayesian bootstrap estimator $\hat{P}_n^*$ is a matrix-beta distribution (product of independent Dirichlet distributions).

Bayesian Bootstrap Algorithm

The steps of the Bayesian bootstrap algorithm are summarized below:

- ① Simulate iid $Z_{it} \sim \text{Exp}(1)$, $i = 1, \dots, m$, $t = 1, \dots, n_i$.
- ② Calculate $\hat{p}_n^*(i, j) := \frac{\sum_{t \in B_{ij}} Z_{it}}{\sum_{t=1}^{n_i} Z_{it}}$ and obtain the matrix estimator $\hat{P}^*$.
- ③ Repeat these steps B times to obtain the estimators $\hat{P}_{n1}^*, \dots, \hat{P}_{nB}^*$ and approximate the posterior distribution $\pi(P \mid x)$ using them.

Central Limit Theorem for the Bayesian Bootstrap

The Bayesian Bootstrap procedure that we developed satisfies a desirable asymptotic property, the Central Limit Theorem.

Theorem (CLT for the Bayesian Bootstrap)

Let $i, j \in S$ and $\hat{p}_{ij}$ denote the maximum likelihood estimator of the transition probability p_{ij} . Then, for almost all sample sequences $\underline{x} = (x_0, x_1, \dots, x_n)$ we have

$$\frac{\sqrt{n_i}(\hat{p}_n^*(i, j) - \hat{p}_{ij})}{\sqrt{\hat{p}_{ij}(1 - \hat{p}_{ij})}} \Big|_{\underline{x}} \xrightarrow{d} \mathcal{N}(0, 1), \quad (6)$$

as $n \rightarrow \infty$. That is, the Bayesian bootstrap estimators are asymptotically normal.

The idea of the proof is to decompose the quantity in (6) in simpler pieces, for which the classical Lindeberg-Lévy Central Limit Theorem can be applied directly.

Proof

Proof. Let $i, j \in S$. We define the quantities

$$S_{nij} := \sum_{t \in B_{ij}} (Z_{it} - 1) \quad \text{and} \quad S_{n_i, n_i - n_{ij}} := \sum_{t=1}^{n_i} (Z_{it} - 1) - \sum_{t \in B_{ij}} (Z_{it} - 1).$$

Since $|B_{ij}| = n_{ij}$, the above quantities can be rewritten as

$$S_{nij} = \sum_{t \in B_{ij}} Z_{it} - n_{ij} \quad \text{and} \quad S_{n_i, n_i - n_{ij}} = \sum_{t=1}^{n_i} Z_{it} - \sum_{t \in B_{ij}} Z_{it} - n_i + n_{ij}.$$

We have that

$$\sqrt{n_i}(\hat{p}_n^*(i, j) - \hat{p}_{ij}) = \sqrt{n_i} \left(\frac{\sum_{t \in B_{ij}} Z_{it}}{n_{ij}} - \frac{n_{ij}}{n_i} \right) = \sqrt{n_i} \cdot \frac{\sum_{t \in B_{ij}} Z_{it}}{n_{ij}} - \sqrt{n_i} \cdot \frac{n_{ij}}{n_i},$$

thus we get that

$$\sqrt{n_i}(\hat{p}_n^*(i, j) - \hat{p}_{ij}) = \sqrt{n_i} \cdot \frac{\sum_{t \in B_{ij}} Z_{it}}{\sum_{t=1}^{n_i} Z_{it}} - \frac{n_{ij}}{\sqrt{n_i}} = \frac{n_i}{\sum_{t=1}^{n_i} Z_{it}} \left(\frac{\sum_{t \in B_{ij}} Z_{it}}{\sqrt{n_i}} - \frac{n_{ij}}{n_i} \cdot \frac{\sum_{t=1}^{n_i} Z_{it}}{\sqrt{n_i}} \right). \quad (7)$$

Notice that

$$\begin{aligned} & \left(1 - \frac{n_{ij}}{n_i}\right) \sqrt{\frac{n_{ij}}{n_i}} \cdot \frac{S_{n_{ij}}}{\sqrt{n_{ij}}} - \frac{n_{ij}}{n_i} \sqrt{1 - \frac{n_{ij}}{n_i}} \cdot \frac{S_{n_i, n_i - n_{ij}}}{\sqrt{n_i - n_{ij}}} \\ &= \left(1 - \frac{n_{ij}}{n_i}\right) \sqrt{\frac{n_{ij}}{n_i}} \cdot \frac{\sum_{t \in B_{ij}} Z_{it} - n_{ij}}{\sqrt{n_{ij}}} - \frac{n_{ij}}{n_i} \sqrt{1 - \frac{n_{ij}}{n_i}} \cdot \frac{\sum_{t=1}^{n_i} Z_{it} - \sum_{t \in B_{ij}} Z_{it} - n_i + n_{ij}}{\sqrt{n_i - n_{ij}}} \\ &= \left(1 - \frac{n_{ij}}{n_i}\right) \frac{\sum_{t \in B_{ij}} Z_{it} - n_{ij}}{\sqrt{n_{ij}}} - \frac{n_{ij}}{n_i} \cdot \frac{n_{ij} - \sum_{t \in B_{ij}} Z_{it}}{\sqrt{n_i}} - \frac{n_{ij}}{n_i} \cdot \frac{\sum_{t=1}^{n_i} Z_{it} - n_i}{\sqrt{n_i}} \end{aligned}$$

$$= \frac{\sum_{t \in B_{ij}} Z_{it}}{\sqrt{n_i}} - \cancel{\frac{n_{ij}}{\sqrt{n_i}}} - \frac{n_{ij}}{n_i} \cdot \frac{\sum_{t=1}^{n_i} Z_{it}}{\sqrt{n_i}} + \cancel{\frac{n_{ij}}{\sqrt{n_i}}} = \frac{\sum_{t \in B_{ij}} Z_{it}}{\sqrt{n_i}} - \frac{n_{ij}}{n_i} \cdot \frac{\sum_{t=1}^{n_i} Z_{it}}{\sqrt{n_i}},$$

thus (7) becomes

$$\begin{aligned} \sqrt{n_i}(\hat{p}_n^*(i, j) - \hat{p}_{ij}) &= \frac{n_i}{\sum_{t=1}^{n_i} Z_{it}} \left[\left(1 - \frac{n_{ij}}{n_i}\right) \sqrt{\frac{n_{ij}}{n_i}} \cdot \frac{S_{n_{ij}}}{\sqrt{n_{ij}}} - \frac{n_{ij}}{n_i} \sqrt{1 - \frac{n_{ij}}{n_i}} \cdot \frac{S_{n_i, n_i - n_{ij}}}{\sqrt{n_i - n_{ij}}} \right] \\ &= \frac{n_i}{\sum_{t=1}^{n_i} Z_{it}} \left[\underbrace{\left(1 - \hat{p}_{ij}\right) \sqrt{\frac{n_{ij}}{n_i}} \cdot \frac{S_{n_{ij}}}{\sqrt{n_{ij}}}}_{(I)} - \underbrace{\hat{p}_{ij} \sqrt{1 - \hat{p}_{ij}} \cdot \frac{S_{n_i, n_i - n_{ij}}}{\sqrt{n_i - n_{ij}}}}_{(II)} \right]. \end{aligned} \quad (8)$$

Since $Z_{it} \sim \text{Exp}(1)$, the Strong Law of Large Numbers yields

$$\frac{1}{n_i} \sum_{t=1}^{n_i} Z_{it} \xrightarrow{\text{a.s.}} \mathbb{E}(Z_{it}) = 1 \Leftrightarrow \frac{n_i}{\sum_{t=1}^{n_i} Z_{it}} \xrightarrow{\text{a.s.}} 1, \quad (9)$$

thus it suffices to prove that the quantities (I), (II) converge in distribution to appropriate normal distributions. We have that

$$\begin{aligned} \frac{(I)}{(1 - \hat{p}_{ij})\sqrt{\hat{p}_{ij}}} &= \frac{(1 - \hat{p}_{ij})\sqrt{\frac{n_{ij}}{n_i}} \cdot \frac{S_{n_{ij}}}{\sqrt{n_{ij}}}}{(1 - \hat{p}_{ij})\sqrt{\hat{p}_{ij}}} = \frac{S_{n_{ij}}}{\sqrt{n_i}\sqrt{\hat{p}_{ij}}} = \frac{S_{n_{ij}}}{\sqrt{n_i}\sqrt{\frac{n_{ij}}{n_i}}} = \frac{S_{n_{ij}}}{\sqrt{n_{ij}}} \\ &= \frac{\sum_{t \in B_{ij}} (Z_{it} - 1)}{\sqrt{n_{ij}}} = \frac{\sum_{t \in B_{ij}} Z_{it} - n_{ij}}{\sqrt{n_{ij}}} \stackrel{|B_{ij}|=n_{ij}}{=} \frac{\sum_{t \in B_{ij}} Z_{it} - \sum_{t \in B_{ij}} \mathbb{E}(Z_{it})}{\sqrt{n_{ij}} \text{Var}(Z_{it})}. \end{aligned}$$

Applying the Lindeberg-Lévy Central Limit Theorem, we get

$$\frac{(I)}{(1 - \hat{p}_{ij})\sqrt{\hat{p}_{ij}}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (10)$$

Using similar calculations, we also get

$$\frac{(II)}{\hat{p}_{ij}\sqrt{1 - \hat{p}_{ij}}} \xrightarrow{d} \mathcal{N}(0, 1). \quad (11)$$

From (10) and (11), we receive that

$$\frac{(I)}{\sqrt{\hat{p}_{ij}(1 - \hat{p}_{ij})}} = \frac{(I)}{(1 - \hat{p}_{ij})\sqrt{\hat{p}_{ij}}} \cdot \sqrt{1 - \hat{p}_{ij}} \xrightarrow{d} \mathcal{N}(0, 1 - \hat{p}_{ij}) \quad (12)$$

and

$$\frac{(II)}{\sqrt{\hat{p}_{ij}(1 - \hat{p}_{ij})}} = \frac{(II)}{\hat{p}_{ij}\sqrt{1 - \hat{p}_{ij}}} \cdot \sqrt{\hat{p}_{ij}} \xrightarrow{d} \mathcal{N}(0, \hat{p}_{ij}). \quad (13)$$

Combining 9, 12 and 13, Slutsky's Lemma yields

$$\frac{\sqrt{n_i}(\hat{p}_n^*(i, j) - \hat{p}_{ij})}{\sqrt{\hat{p}_{ij}(1 - \hat{p}_{ij})}} \Big|_{\underline{x}} = \frac{n_i}{\sum_{t=1}^{n_i} Z_{it}} \cdot \left[\frac{(I)}{\sqrt{\hat{p}_{ij}(1 - \hat{p}_{ij})}} - \frac{(II)}{\sqrt{\hat{p}_{ij}(1 - \hat{p}_{ij})}} \right] \xrightarrow{d} \mathcal{N}(0, 1)$$

and the proof is complete. □

Application

- We compare the classical asymptotic approach with the Bayesian bootstrap approach in the problem of estimating transition probabilities.
- The goal is to estimate the probability p_{12} of the transition $1 \rightarrow 2$ from a Markov chain path.
- We will not use real data, but rather simulate a path from a Markov chain with transition probability matrix

$$P = \begin{pmatrix} 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \\ 0.4 & 0.4 & 0.2 \end{pmatrix}.$$

- We compare the two matrix estimators and the confidence intervals for p_{12} .
- We do so for two sample sizes: $N = 50$ and $N = 100$.

$N = 50$

- We generate a path of length 50 according to the matrix P .
- P_{asym} denotes the classical maximum likelihood estimator given by (2).
- P_{boot} denotes the Bayesian bootstrap estimator given by (5).
- We get the matrix estimators

$$P_{asym} = \begin{pmatrix} 0.2941176 & 0.4117647 & 0.2941176 \\ 0.1875000 & 0.2500000 & 0.5625000 \\ 0.5625000 & 0.3125000 & 0.1250000 \end{pmatrix}$$

$$P_{boot} = \begin{pmatrix} 0.2455232 & 0.3522173 & 0.40225948 \\ 0.1665064 & 0.3294817 & 0.50401188 \\ 0.6290134 & 0.3524531 & 0.01853348 \end{pmatrix},$$

where P_{boot} is the matrix from one bootstrap replication.

- Notice that they are somewhat close to each other.

Asymptotic Confidence Interval

Proposition

Let $I(p_{ij})$ denote the $100(1 - a)\%$ -confidence interval for p_{ij} . Then,

$$I(p_{ij}) = \left(\hat{p}_{ij}(n) - z_{a/2} \sqrt{\frac{\hat{p}_{ij}(n)(1 - \hat{p}_{ij}(n))}{n_i(n)}}, \hat{p}_{ij}(n) + z_{a/2} \sqrt{\frac{\hat{p}_{ij}(n)(1 - \hat{p}_{ij}(n))}{n_i(n)}} \right).$$

In our sample we find the maximum likelihood estimator $\hat{p}_{12}(50) = 0.4117647$. Using the above proposition, we find that the 95%-confidence interval for p_{12} is

$$I_{asym}(p_{12}) = (0.1778143, 0.6457151).$$

Bayesian Bootstrap Confidence Interval

We saw that the Bayesian bootstrap distribution of each row is Dirichlet, i.e.

$$\hat{P}_{i\cdot} \sim \text{Dir}(n_{i1}, n_{i2}, n_{i3}).$$

We will use the following Proposition to find the distribution of p_{12} .

Proposition

If $X = (X_1, \dots, X_m) \sim \text{Dir}(c_1, \dots, c_m)$, then the marginal distribution of X_k is

$$X_k \sim \text{Beta}(c_k, c_1 + \dots + c_{k-1} + c_{k+1} + \dots + c_m), \quad \forall k \in [m].$$

In our scenario, we have $\hat{p}_{12}^* \sim \text{Beta}(n_{12}, n_{11} + n_{13}) \equiv \text{Beta}(7, 10)$. Using the quantiles of the Beta distribution, we can get the 95% HPD-region

$$I_{bb}(p_{12}) = (0.1975341, 0.6456539),$$

which has smaller length than the asymptotic confidence interval.

$$N = 100$$

We do the same for $N = 100$. We get the following results:

$$P_{asym} = \begin{pmatrix} 0.3684211 & 0.3684211 & 0.26315789 \\ 0.1935484 & 0.2258065 & 0.58064516 \\ 0.6000000 & 0.3333333 & 0.06666667 \end{pmatrix}$$

$$P_{boot} = \begin{pmatrix} 0.36953846 & 0.3535368 & 0.2769248 \\ 0.08861432 & 0.3139951 & 0.5973906 \\ 0.53312820 & 0.3107638 & 0.1561080 \end{pmatrix}$$

$$I_{asym}(p_{12}) = (0.2150504, 0.5217917)$$

$$I_{bb}(p_{12}) = (0.2245761, 0.5253887)$$

Remark

In both cases ($N = 50$ and $N = 100$), the matrix estimators of the two methods are close to each other, while the length of the Bayesian bootstrap credible region is smaller than the length of the asymptotic confidence interval.

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